

Ayoub ADIL

Software engineer, *in 3D and beyond.*

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I design, develop and maintain software applications, from the first architecture to support after delivery. I work on 3D software, such as point clouds, 3D viewers and reconstruction, and on larger systems: multi-service platforms and the applications their users work with. Several of my applications run in production today.

EXPERIENCE

R&D Vision — 03/2024 – Present

Software Engineer · Same role, renamed when the company specialised its job titles — 09/2025 – Present

Application Engineer — 09/2024 – 09/2025

R&D Intern · Master's final internship: 3D reconstruction from images — 03/2024 – 08/2024

Depending on the project, I work solo, in a pair or in a larger team, and I install and support what I deliver.

INDUSTRIAL PROJECTS · 2024 – PRESENT

Tunnel inspection platform — In progress · 2026

Rail tunnel operator. Developed most of the services and the communication between them; Developed all the applications: operator tablet, multi-screen inspection, history comparison.

`.NET · WPF · C# · C++ · Modbus / JBus · Phoxene · NI DAQ`

LiDAR safety barrier for e-bus high-voltage charging — Proof of concept · 2026

Transport-sector manufacturer. Wrote the point-cloud processing that tells real intrusions from rain, snow and bus motion; Integrated and calibrated the LiDARs; Built the operator interface.

`Python · C++ · C# · WPF / .NET`

3D dough monitoring in an industrial oven — In production · 2024

Food-industry group. Integrated the ToF camera; Built the 3D viewer and the depth-image display; Built the operator interface. `C++`

Track inspection software, 3 vehicle configurations — In production · 2026

Rail inspection company. Took over a colleague's code and fixed malfunctions; Refactored it into one modular application for the 3 configurations; Installed it on trains; Support and maintenance.

`.NET · WPF · C#`

Signal-box route-setting assistant — In production · 2025

Rail operator. Built the complete application; Recording, history and replay mode.

`.NET · WPF · C# · C++`

Fabric defect inspection — Delivered · 2025 – 2026

Luxury fashion house. Built the operator interface; Built the dataset-creation tools.

`C# · C++ · WPF / .NET`

Train passage detection — Delivered · 2026

Rail operator. Built the data-browsing application; Built the data service. `.NET · WPF`

Metro tunnel niche detection — In production · 2024

Urban transit operator. Took over a colleague's code; Finalised the niche-detection application.

`WPF`

MASTER'S INTERNSHIP & ACADEMIC PROJECTS

3D reconstruction of rail and metro tunnels from images · Master's internship · 03/2024 – 08/2024
Reviewed the visual-SLAM literature; Implemented and ran ORB-SLAM and PLVS; Applied them to real rail and metro footage and analysed the results. `C++ · Docker · Ubuntu`

Monte Carlo path tracing · Master course project · 12/2023 `C++ · OpenGL · Shaders`

Experimental protocols in virtual reality · Master project · ICube lab · 09/2023 – 12/2023
`Unity · VR`

Interaction with objects in virtual reality · Research project · ICube lab · 01/2023 – 06/2023
`Unreal Engine 5 · VR`

Smart home · Bachelor's internship · FST Mohammedia · 04/2020 – 06/2020 `Node.js · Python · TypeScript`

EDUCATION

2021 – 2024
Master's, Image & 3D
UFR Math-Info · University of Strasbourg

2020 – 2021
Bachelor's (final year), Computer Science
INP-ISIMA · Clermont Auvergne University

2019 – 2020
Bachelor's, Computer Science, Networks & Multimedia
FST Mohammedia · Hassan II University

2017 – 2019
DEUG: Mathematics, Computer Science & Physics
FST Mohammedia · Hassan II University

2016 – 2017
Baccalaureate, Mathematical Sciences
Lycée Ben Msik

SKILLS

LANGUAGES
C++, C#, Python, TypeScript, JavaScript

3D & GRAPHICS
OpenGL / shaders, three.js, Unity, Unreal Engine 5, Blender

VISION & DATA
OpenCV, Point clouds, Visual SLAM, Stereo calibration

FRAMEWORKS
.NET / WPF, React, Node.js

TOOLS
Git, Docker, Linux

HARDWARE
Line-scan cameras, Area-scan cameras, ToF cameras, Thermal cameras, LiDAR, NI DAQ, Modbus / JBus, RFID / GPS

LANGUAGES

Arabic	Native
French	Fluent
English	Conversational
German	Learning